SEQUENCE 9

Proportions and percentage change

21–29 January 2027

9.1 Proportion and proportion of a proportion

Key point · Definitions. In a nonempty finite set E, the proportion of a subset F is $p = \text{Card}(F)/\text{Card}(E)$; $0 \leq p \leq 1$. Its expression as a percentage is $100p\,\%$. If $G \subseteq F \subseteq E$, $F \neq \varnothing$, and if p_1 is the proportion of F in E and p_2 that of G in F, the proportion of G in E is $p_1 p_2$.

Example 1. A school has 1500 students, of whom 652 are day boarders. Calculate their proportion. In a class, 60% of the students play sport; among them, 40% are boys. What proportion of the class do the boys who play sport represent?

Example 2. In a fictional school of 800 students, 35% play a sport at a club; among them, 45% play a team sport. Calculate the two counts and the final proportion.

Watch out. Proportion and rate of change do not answer the same question: a proportion compares a part to a whole, a rate compares a change to an initial value.

9.2 Absolute change and rate of change

Key point · Definition. For an initial value $V_I > 0$ and a final value $V_F > 0$, the absolute change is $V_F - V_I$; the rate of change is

$$t = \frac{V_F - V_I}{V_I}.$$

The rate has no unit. It is positive for an increase, negative for a decrease. The reference is always the initial value.

Example 3. A cheese is sold for 2.13 euros instead of 3.55 euros. Calculate the absolute change and the rate.

Key point · Multiplier. $CM = V_F/V_I = 1+t$, so $V_F = V_I(1+t)$. To recover V_I, divide V_F by $1+t$ (nonzero).

Example 4. A fictional item priced 50000 Ar increases by 12%, then separately consider a 12% discount on the initial price. Give the final prices. A price after a 20% increase is 72000 Ar; find its initial price.

Fluency · 5–10 min. Expand $(x-3)^2$; solve $1/x = 5$; give the mean of 8; 10; 12.

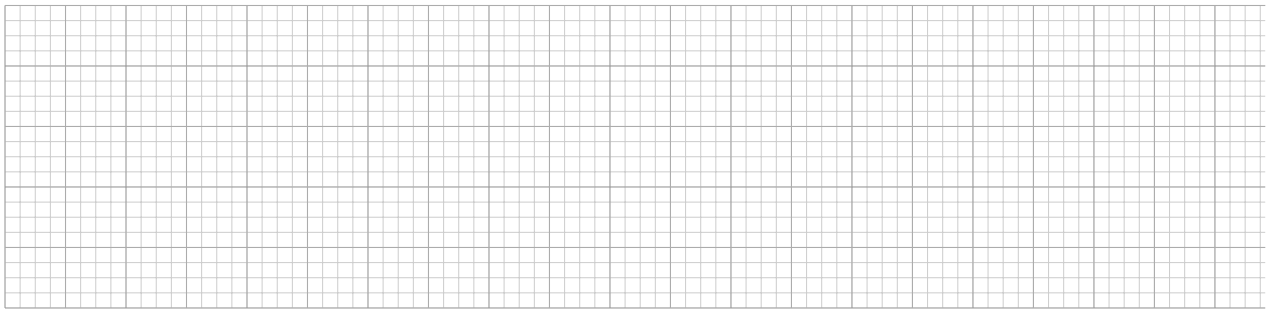
9.3 Successive changes

Key point · Property. Successive changes with rates t_1, t_2 have overall multiplier

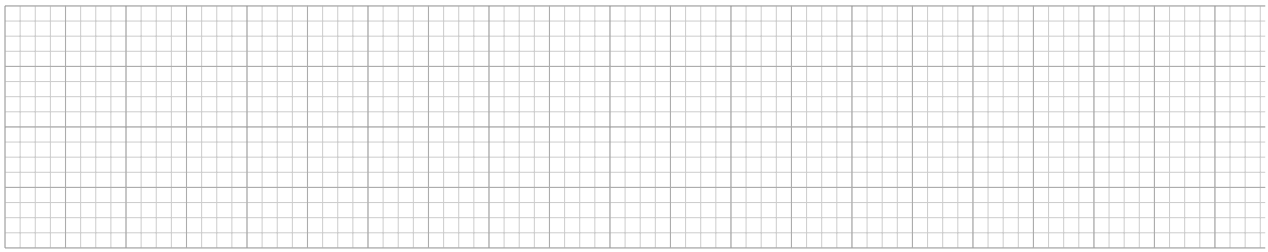
$$CM_G = (1 + t_1)(1 + t_2), \qquad t_G = (1 + t_1)(1 + t_2) - 1.$$

We multiply the multipliers; we do not simply add the rates.

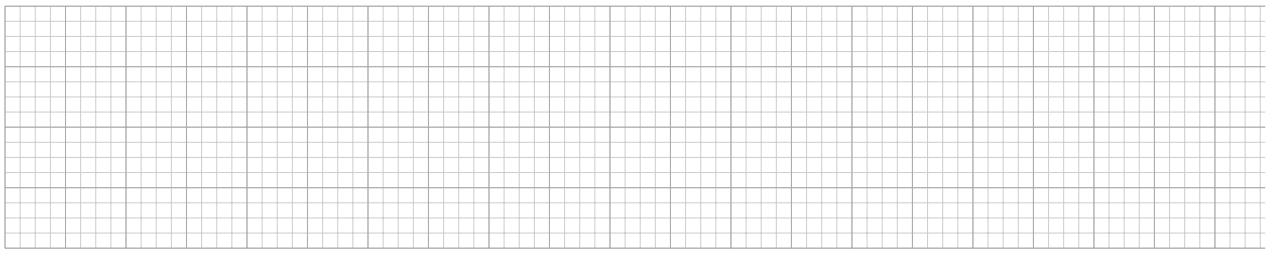
Example 5. A manufacturer sold 5600 bicycles in 2022, which is 25% more than in 2021. Its sales decrease by 15% in 2023. Determine the sales of 2021 and 2023, then the overall rate between these two years.



Example 6. A price increases by 10% then decreases by 10%. Does it return to the initial price?



Example 7. A rate goes from 20% to 25%. Explain the difference between the increase in percentage points and the relative change.

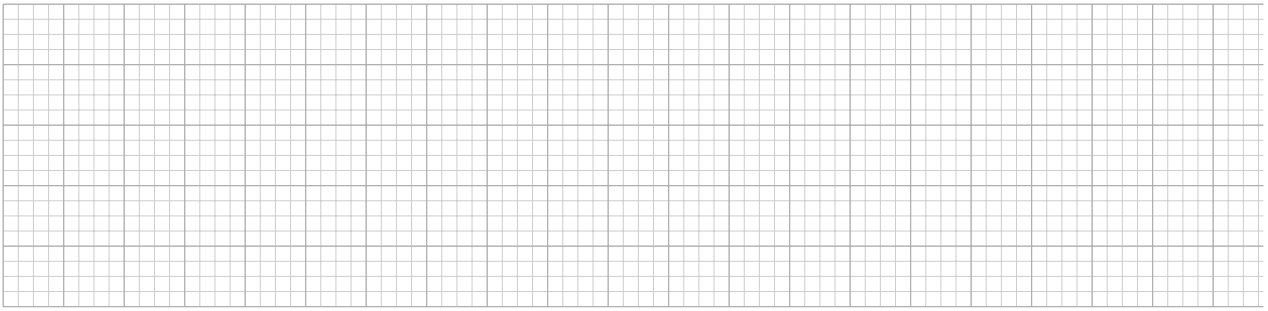
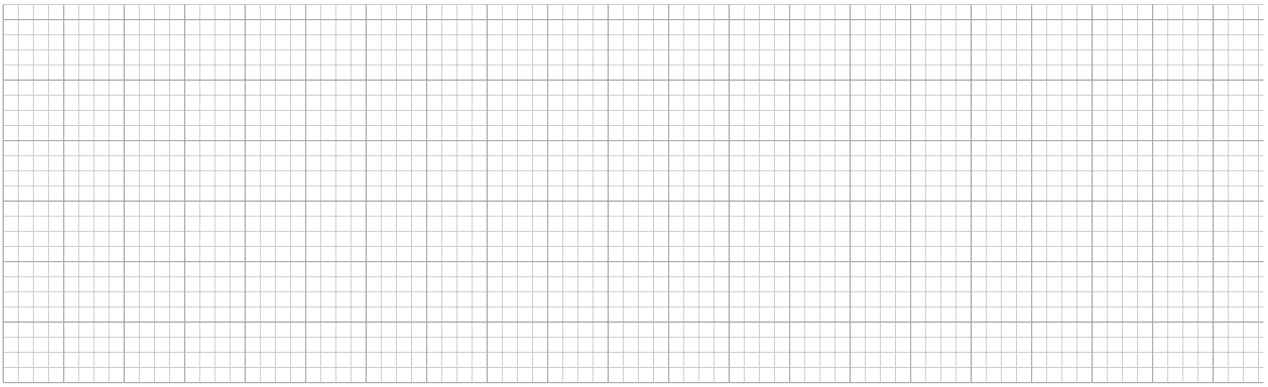


9.4 Reverse change

Key point · Property. To return to the initial value after a change of rate $t > -1$, use the multiplier $1/(1+t)$. The reverse rate is

$$t_{\text{reverse}} = \frac{1}{1+t} - 1 = \frac{-t}{1+t}.$$

Example 8. Which rate compensates a 25% increase? Which rate compensates a 20% decrease?



Example 9. A price undergoes a 30% discount, then a 10% increase. What additional percentage increase restores the initial price?